

Worked example: IAP × Sports interaction

Outcome 1 — Install success | Pooled analysis

1 Construct the two exposure-by-outcome tables

Sports stratum (n = 136,416)

	IAP = 1	IAP = 0	Total
Success	21,101	16,214	37,315
Not success	54,664	44,437	99,101
Total	75,765	60,651	136,416

Non-Sports stratum (n = 1,642,284)

	IAP = 1	IAP = 0	Total
Success	403,136	449,203	852,339
Not success	442,915	347,030	789,945
Total	846,051	796,233	1,642,284

2 Estimate conditional success probabilities

Sports

Outcome	IAP = 1	IAP = 0
Success	21,101 / 75,765 = 0.2785	16,214 / 60,651 = 0.2673

Non-Sports

Outcome	IAP = 1	IAP = 0
Success	403,136 / 846,051 = 0.4765	449,203 / 796,233 = 0.5642

$$\Delta p_{\text{Sports}} = 0.2785 - 0.2673 = +0.0112$$

$$OR_{\text{Sports}} = \frac{21101 \times 44437}{54664 \times 16214} = 1.0579$$

$$\Delta p_{\text{Non-Sports}} = 0.4765 - 0.5642 = -0.0877$$

$$OR_{\text{Non-Sports}} = \frac{403136 \times 347030}{442915 \times 449203} = 0.7032$$

Pooled interaction analysis

Install success | Additive and multiplicative scales

3 Additive interaction

$$DDP = \Delta p_{\text{Sports}} - \Delta p_{\text{Non-Sports}}$$

$$= +0.0112 - (-0.0877) = +0.0988$$

95% CI: [0.0939, 0.1038]

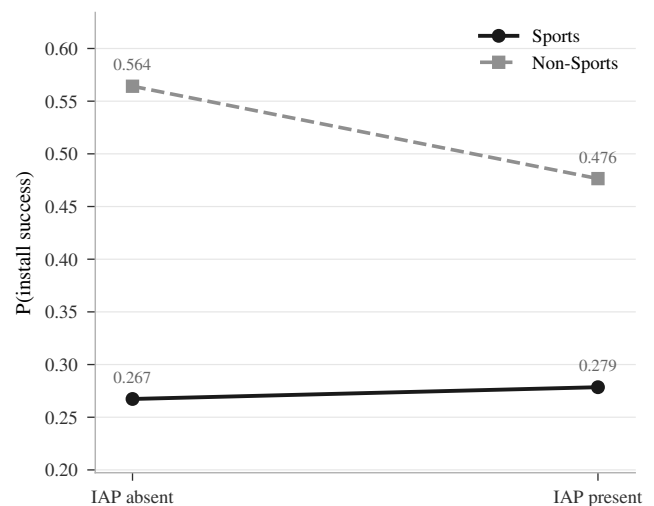
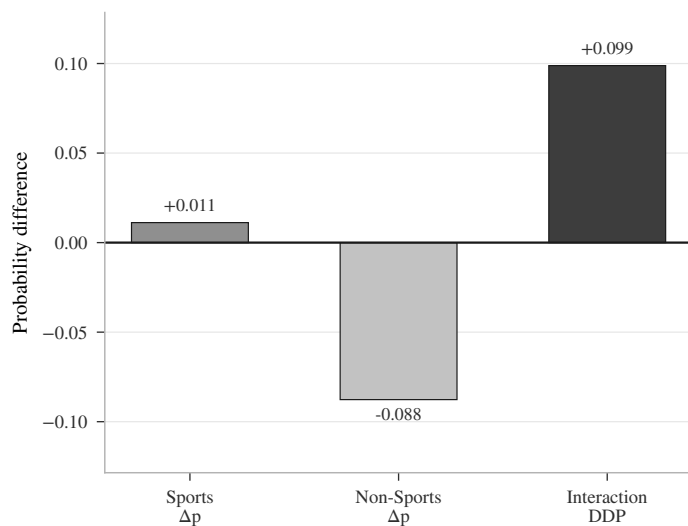


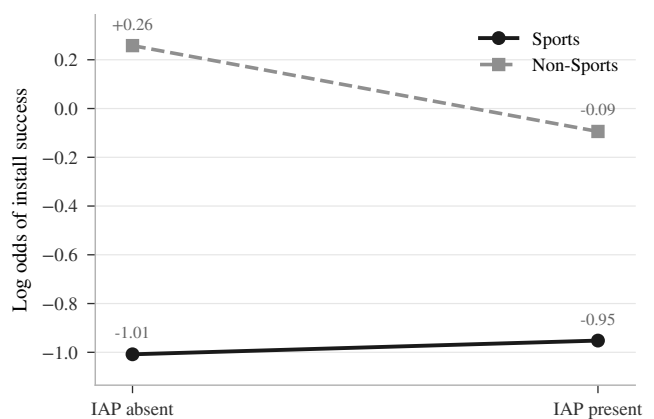
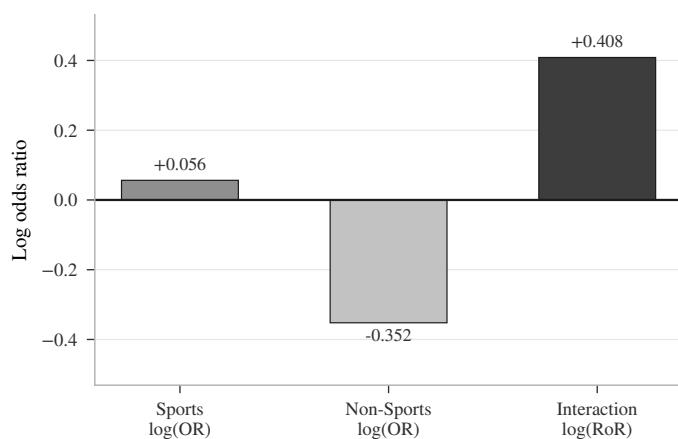
Figure 1. Additive interaction. The left panel decomposes the two within-stratum probability differences and their difference-in-differences; the right panel shows the corresponding conditional success probabilities.

4 Multiplicative interaction

$$RoR = OR_{\text{Sports}} / OR_{\text{Non-Sports}}$$

$$= 1.0579 / 0.7032 = 1.5045$$

95% CI: [1.4677, 1.5423]



Cross-market replication

Install success | United States and India

Country-specific calculations

United States

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	471	1,644	298	835
Non-Sports	11,732	11,890	8,681	6,799

$$\Delta p_{\text{Sports}} = 0.2227 - 0.2630 = -0.0403$$

$$\Delta p_{\text{Non-Sports}} = 0.4967 - 0.5608 = -0.0641$$

$$\text{DDP} = -0.0403 - (-0.0641) = \mathbf{+0.0238}$$

$$\text{RoR} = 0.8028/0.7728 = \mathbf{1.0388} \qquad \text{Joint: } \mathbf{00}$$

India

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	691	452	513	1,592
Non-Sports	4,875	7,641	15,103	11,483

$$\Delta p_{\text{Sports}} = 0.6045 - 0.2437 = +0.3608$$

$$\Delta p_{\text{Non-Sports}} = 0.3895 - 0.5681 = -0.1786$$

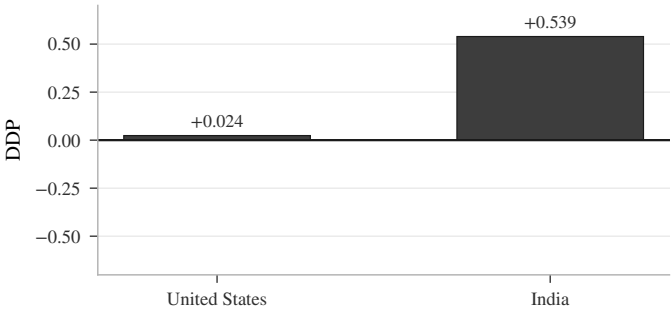
$$\text{DDP} = +0.3608 - (-0.1786) = \mathbf{+0.5394}$$

$$\text{RoR} = 4.7442/0.4851 = \mathbf{9.7802} \qquad \text{Joint: } \mathbf{++}$$

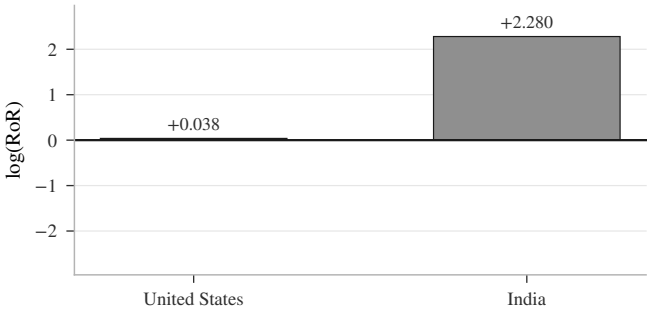
Comparison of interaction estimates

Context	Sports Δp	Non-Sports Δp	DDP	Sports OR	Non-Sports OR	RoR	Joint
United States	-0.040	-0.064	+0.024	0.803	0.773	1.039	00
India	+0.361	-0.179	+0.539	4.744	0.485	9.780	++

Additive interaction



Multiplicative interaction



Temporal replication

Install success | Three United States monthly contexts

Selected monthly contexts

Jan 2025

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	54	105	14	59
Non-Sports	819	904	626	444

$$\Delta p_{\text{Sports}} = 0.3396 - 0.1918 = +0.1478$$

$$\Delta p_{\text{Non-Sports}} = 0.4753 - 0.5850 = -0.1097$$

$$\text{DDP} = +0.1478 - (-0.1097) = \mathbf{+0.2576}$$

$$\text{RoR} = 2.1673/0.6426 = \mathbf{3.3729}$$

Joint: ++

Aug 2025

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	14	129	22	67
Non-Sports	834	844	643	472

$$\Delta p_{\text{Sports}} = 0.0979 - 0.2472 = -0.1493$$

$$\Delta p_{\text{Non-Sports}} = 0.4970 - 0.5767 = -0.0797$$

$$\text{DDP} = -0.1493 - (-0.0797) = \mathbf{-0.0696}$$

$$\text{RoR} = 0.3305/0.7254 = \mathbf{0.4557}$$

Joint: 0-

Feb 2026

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	49	88	28	67
Non-Sports	831	854	605	503

$$\Delta p_{\text{Sports}} = 0.3577 - 0.2947 = +0.0629$$

$$\Delta p_{\text{Non-Sports}} = 0.4932 - 0.5460 = -0.0529$$

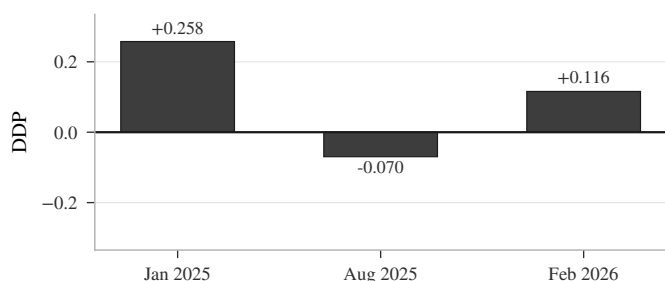
$$\text{DDP} = +0.0629 - (-0.0529) = \mathbf{+0.1158}$$

$$\text{RoR} = 1.3324/0.8090 = \mathbf{1.6469}$$

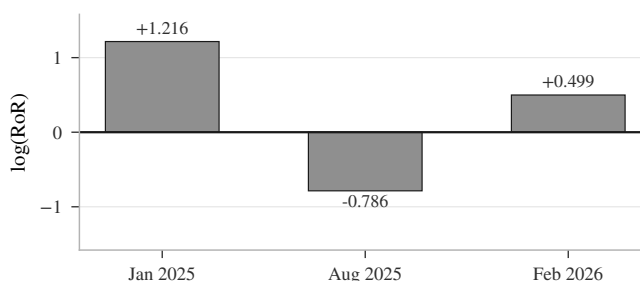
Joint: 00

Temporal comparison

Additive interaction



Multiplicative interaction



Install-success synthesis

Pooled, national, and temporal contexts

Interaction estimates

Context	Sports Δp	Non-Sports Δp	DDP	Sports OR	Non-Sports OR	RoR	Joint
Pooled	+0.011	-0.088	+0.099	1.058	0.703	1.505	++
United States	-0.040	-0.064	+0.024	0.803	0.773	1.039	00
India	+0.361	-0.179	+0.539	4.744	0.485	9.780	++
US Jan 2025	+0.148	-0.110	+0.258	2.167	0.643	3.373	++
US Aug 2025	-0.149	-0.080	-0.070	0.331	0.725	0.456	0-
US Feb 2026	+0.063	-0.053	+0.116	1.332	0.809	1.647	00

Interaction magnitude across contexts

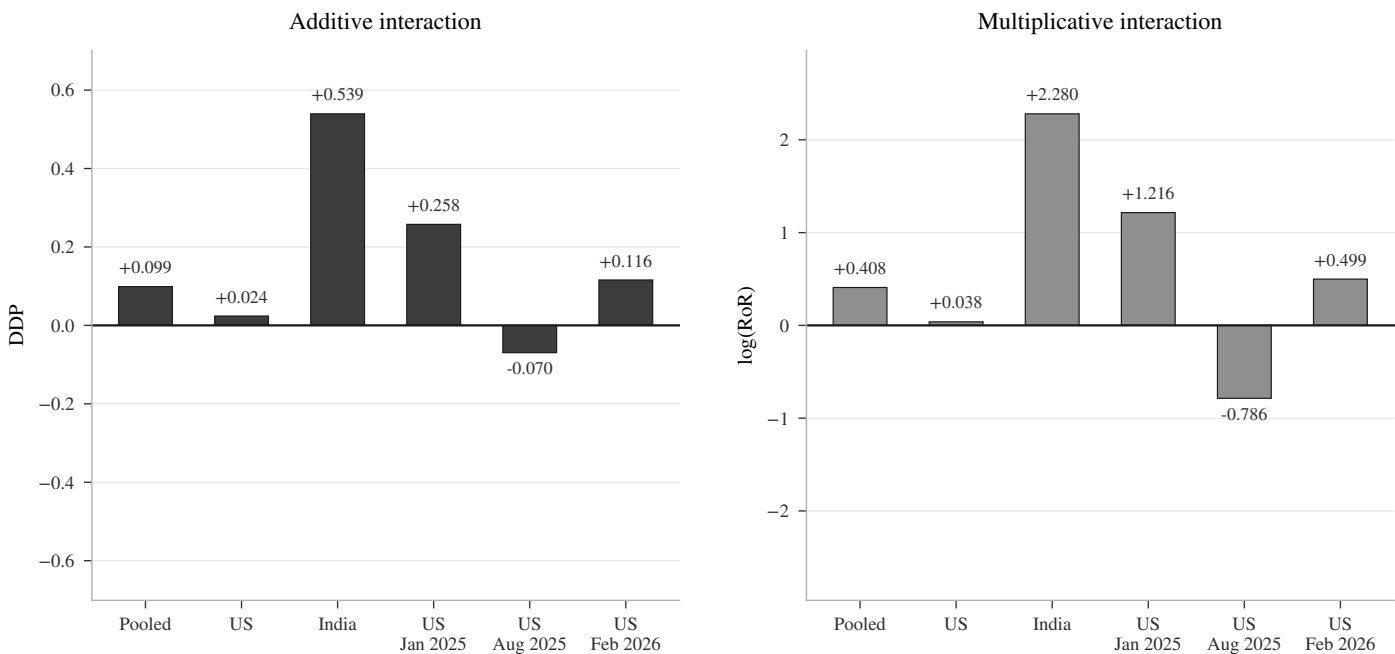


Figure 2. Install-success interaction estimates across the pooled analysis, two national markets, and three United States monthly contexts. Both panels use a zero null: DDP on the additive scale and log(RoR) on the multiplicative scale.

The worked contexts are presented to demonstrate how the same interaction calculation is propagated across analytical levels. The table retains RoR on its original ratio scale for reporting, whereas the figure uses log(RoR) so that the multiplicative null is centered at zero.

Parallel outcome analysis

Outcome 2 — Rating success | Pooled analysis

1 Construct the two exposure-by-outcome tables

Sports				Non-Sports			
	IAP = 1	IAP = 0	Total		IAP = 1	IAP = 0	Total
Success	32,302	26,177	58,479	Success	433,843	408,743	842,586
Not success	43,463	34,474	77,937	Not success	412,208	387,490	799,698
Total	75,765	60,651	136,416	Total	846,051	796,233	1,642,284

2 Estimate within-stratum associations

Stratum	P(success IAP = 1)	P(success IAP = 0)	Δp	OR
Sports	0.4263	0.4316	-0.0053	0.9788
Non-Sports	0.5128	0.5133	-0.0006	0.9978

3 Estimate interaction on both scales

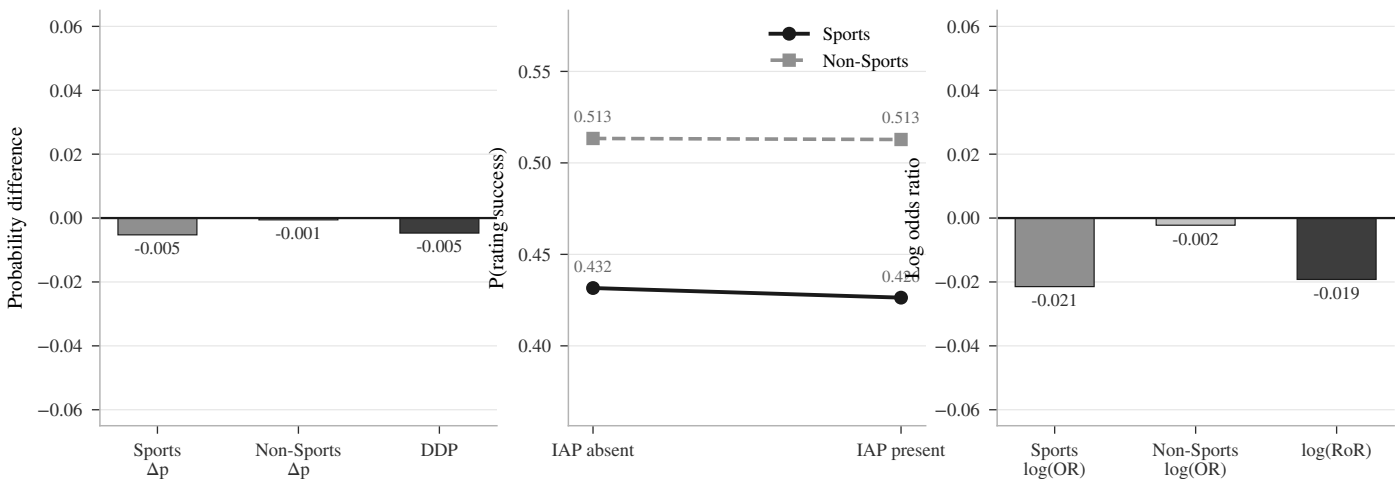
$$\text{DDP} = -0.0053 - (-0.0006) = \mathbf{-0.0047}$$

$$\text{RoR} = 0.9788/0.9978 = \mathbf{0.9810}$$

$$95\% \text{ CI: } [-0.0102, 0.0008]$$

$$95\% \text{ CI: } [0.9592, 1.0032]$$

Joint classification: 00



Rating-success replication

One national context and one monthly context

Context-specific calculations

United States

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	896	1,219	489	644
Non-Sports	12,254	11,368	7,848	7,632

$$\Delta p_{\text{Sports}} = 0.4236 - 0.4316 = -0.0080$$

$$\Delta p_{\text{Non-Sports}} = 0.5188 - 0.5070 = +0.0118$$

$$\text{DDP} = -0.0080 - (+0.0118) = -\mathbf{0.0197}$$

$$\text{RoR} = 0.9680/1.0483 = \mathbf{0.9234} \qquad \text{Joint: } \mathbf{00}$$

United States — Aug 2025

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
Sports	49	94	42	47
Non-Sports	890	788	546	569

$$\Delta p_{\text{Sports}} = 0.3427 - 0.4719 = -0.1293$$

$$\Delta p_{\text{Non-Sports}} = 0.5304 - 0.4897 = +0.0407$$

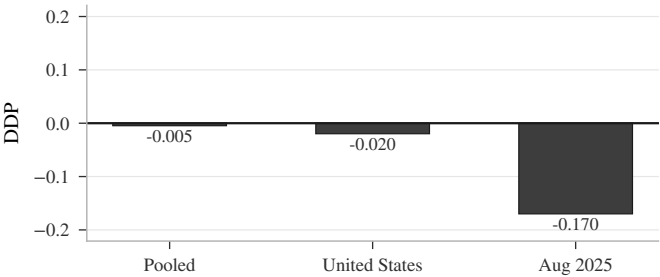
$$\text{DDP} = -0.1293 - (+0.0407) = -\mathbf{0.1700}$$

$$\text{RoR} = 0.5833/1.1770 = \mathbf{0.4956} \qquad \text{Joint: } \mathbf{--}$$

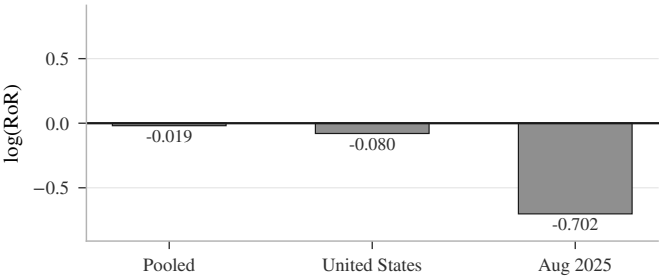
Rating-success summary

Context	Sports Δp	Non-Sports Δp	DDP	Sports OR	Non-Sports OR	RoR	Joint
Pooled	-0.005	-0.001	-0.005	0.979	0.998	0.981	00
United States	-0.008	+0.012	-0.020	0.968	1.048	0.923	00
US Aug 2025	-0.129	+0.041	-0.170	0.583	1.177	0.496	--

Additive interaction



Multiplicative interaction



Complete IAP × Sports interaction summary

Pooled, cross-market, and temporal evidence

Interaction estimates across analytical levels

Outcome	Analytical level	Mean / pooled DDP	95% CI / observed range	Mean / pooled RoR	Mean / pooled log(RoR)	++	--	00	Mixed	N
Install	Pooled	+0.099	[+0.094, +0.104]	1.505	+0.408	1	0	0	0	1
Install	Countries	+0.129	-0.143 to +0.539	2.420	+0.459	23	10	6	3	42
Install	Country × month	+0.128	-0.305 to +0.815	2.921	+0.361	242	85	217	44	588
Rating	Pooled	-0.005	[-0.010, +0.001]	0.981	-0.019	0	0	1	0	1
Rating	Countries	+0.004	-0.138 to +0.136	1.048	+0.015	13	14	15	0	42
Rating	Country × month	+0.004	-0.276 to +0.329	1.102	+0.013	57	51	479	1	588

Pooled rows report the interaction estimate and its 95% confidence interval. Country and country × month rows summarize the complete set of context-specific estimates; the interval column therefore shows the observed range rather than a confidence interval.

Cross-market classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Countries
Install success	+0.129	2.420	23	10	6	3	42
Rating success	+0.004	1.048	13	14	15	0	42

Country × month classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Contexts
Install success	+0.128	2.921	242	85	217	44	588
Rating success	+0.004	1.102	57	51	479	1	588

Joint classification combines additive and multiplicative inference: ++ indicates statistically positive interaction on both scales; -- indicates statistically negative interaction on both scales; 00 indicates null interaction on both scales; Mixed contains all remaining combinations. Multiplicative effects are reported as RoR in the tables and transformed to log(RoR) when a zero-centered visual scale is required.

Country-specific interaction estimates

IAP × Sports interaction | All national markets

Install and rating interaction estimates by country

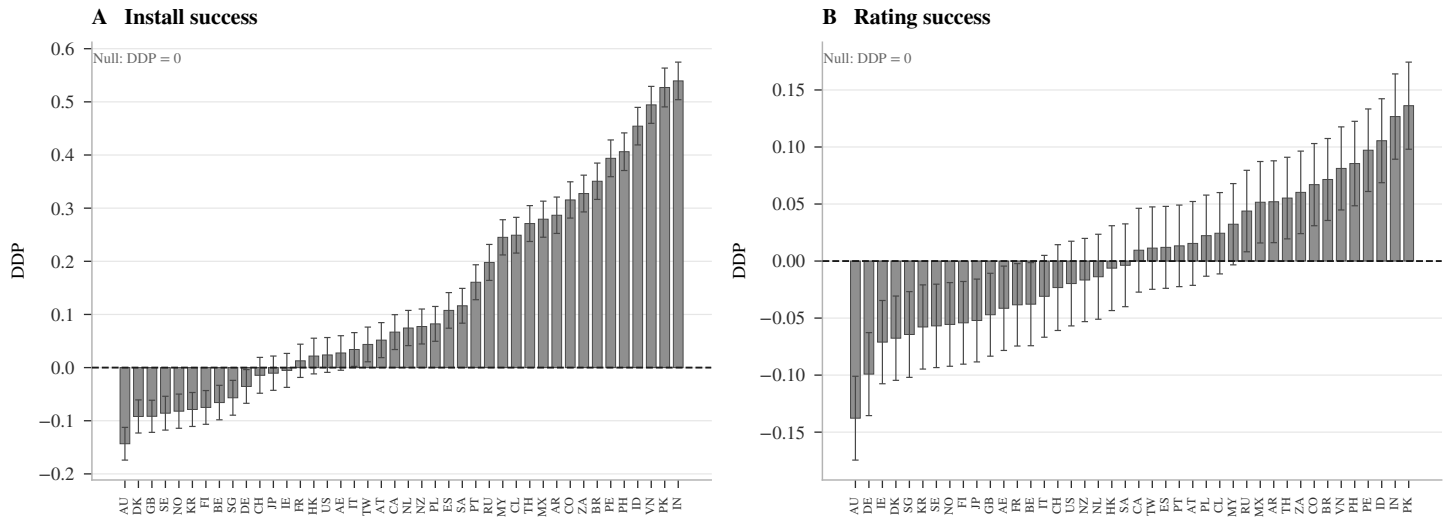
Country	Install DDP [95% CI]	Install RoR [95% CI]	Rating DDP [95% CI]	Rating RoR [95% CI]
AE	+0.028 [-0.005, +0.060]	1.014 [0.851, 1.209]	-0.041 [-0.078, -0.004]	0.842 [0.722, 0.981]
AR	+0.287 [+0.252, +0.321]	3.404 [2.922, 3.964]	+0.052 [+0.016, +0.088]	1.233 [1.067, 1.425]
AT	+0.052 [+0.019, +0.085]	1.208 [1.020, 1.430]	+0.015 [-0.021, +0.052]	1.067 [0.917, 1.240]
AU	-0.143 [-0.174, -0.112]	0.303 [0.248, 0.370]	-0.138 [-0.175, -0.101]	0.564 [0.485, 0.657]
BE	-0.066 [-0.098, -0.033]	0.614 [0.518, 0.729]	-0.038 [-0.074, -0.001]	0.855 [0.737, 0.993]
BR	+0.351 [+0.316, +0.385]	4.490 [3.850, 5.236]	+0.072 [+0.036, +0.107]	1.334 [1.154, 1.542]
CA	+0.067 [+0.034, +0.100]	1.333 [1.125, 1.579]	+0.010 [-0.027, +0.046]	1.040 [0.895, 1.210]
CH	-0.015 [-0.048, +0.019]	0.840 [0.706, 1.000]	-0.023 [-0.061, +0.014]	0.909 [0.778, 1.061]
CL	+0.249 [+0.215, +0.283]	2.933 [2.515, 3.419]	+0.024 [-0.011, +0.060]	1.103 [0.956, 1.274]
CO	+0.315 [+0.281, +0.350]	3.940 [3.375, 4.601]	+0.067 [+0.031, +0.103]	1.310 [1.133, 1.514]
DE	-0.035 [-0.067, -0.004]	0.733 [0.616, 0.872]	-0.099 [-0.136, -0.063]	0.666 [0.574, 0.773]
DK	-0.092 [-0.123, -0.061]	0.476 [0.394, 0.576]	-0.068 [-0.105, -0.031]	0.755 [0.648, 0.881]
ES	+0.108 [+0.074, +0.141]	1.563 [1.337, 1.826]	+0.012 [-0.024, +0.048]	1.049 [0.907, 1.213]
FI	-0.075 [-0.107, -0.043]	0.569 [0.478, 0.678]	-0.054 [-0.090, -0.018]	0.801 [0.690, 0.929]
FR	+0.013 [-0.019, +0.044]	0.947 [0.796, 1.127]	-0.038 [-0.074, -0.002]	0.854 [0.736, 0.992]
GB	-0.092 [-0.122, -0.061]	0.464 [0.383, 0.561]	-0.047 [-0.083, -0.011]	0.823 [0.708, 0.956]
HK	+0.022 [-0.012, +0.055]	1.054 [0.889, 1.250]	-0.006 [-0.043, +0.031]	0.976 [0.837, 1.137]
ID	+0.454 [+0.419, +0.490]	6.762 [5.786, 7.903]	+0.106 [+0.069, +0.142]	1.530 [1.319, 1.775]
IE	-0.005 [-0.037, +0.027]	0.881 [0.741, 1.047]	-0.071 [-0.108, -0.035]	0.747 [0.643, 0.868]
IN	+0.539 [+0.504, +0.575]	9.780 [8.328, 11.486]	+0.127 [+0.089, +0.164]	1.662 [1.430, 1.933]
IT	+0.034 [+0.002, +0.066]	1.077 [0.913, 1.271]	-0.031 [-0.067, +0.005]	0.882 [0.762, 1.021]
JP	-0.011 [-0.043, +0.022]	0.865 [0.731, 1.024]	-0.052 [-0.088, -0.016]	0.808 [0.696, 0.937]
KR	-0.079 [-0.111, -0.047]	0.564 [0.472, 0.673]	-0.058 [-0.095, -0.021]	0.790 [0.680, 0.918]
MX	+0.279 [+0.245, +0.313]	3.302 [2.833, 3.847]	+0.052 [+0.016, +0.087]	1.232 [1.066, 1.424]
MY	+0.245 [+0.212, +0.278]	2.910 [2.490, 3.402]	+0.032 [-0.003, +0.068]	1.139 [0.986, 1.315]
NL	+0.074 [+0.041, +0.108]	1.296 [1.094, 1.536]	-0.014 [-0.051, +0.023]	0.945 [0.813, 1.099]
NO	-0.082 [-0.114, -0.050]	0.545 [0.456, 0.652]	-0.056 [-0.092, -0.019]	0.795 [0.682, 0.926]
NZ	+0.077 [+0.044, +0.110]	1.383 [1.172, 1.632]	-0.017 [-0.053, +0.020]	0.935 [0.806, 1.085]
PE	+0.394 [+0.359, +0.428]	5.334 [4.571, 6.224]	+0.097 [+0.061, +0.133]	1.479 [1.278, 1.712]
PH	+0.406 [+0.371, +0.442]	5.772 [4.924, 6.766]	+0.085 [+0.048, +0.123]	1.414 [1.217, 1.643]
PK	+0.527 [+0.490, +0.564]	9.408 [7.997, 11.067]	+0.136 [+0.098, +0.174]	1.729 [1.482, 2.017]
PL	+0.082 [+0.049, +0.115]	1.421 [1.214, 1.664]	+0.022 [-0.013, +0.058]	1.095 [0.947, 1.266]
PT	+0.161 [+0.128, +0.193]	2.028 [1.729, 2.377]	+0.013 [-0.022, +0.049]	1.056 [0.913, 1.222]
RU	+0.198 [+0.164, +0.232]	2.300 [1.976, 2.678]	+0.044 [+0.008, +0.080]	1.192 [1.033, 1.377]
SA	+0.116 [+0.084, +0.149]	1.664 [1.411, 1.962]	-0.004 [-0.040, +0.033]	0.986 [0.850, 1.143]
SE	-0.086 [-0.118, -0.054]	0.515 [0.430, 0.617]	-0.057 [-0.093, -0.020]	0.790 [0.679, 0.919]
SG	-0.057 [-0.090, -0.024]	0.644 [0.538, 0.772]	-0.064 [-0.102, -0.027]	0.767 [0.658, 0.895]
TH	+0.271 [+0.237, +0.305]	3.210 [2.753, 3.744]	+0.055 [+0.019, +0.091]	1.249 [1.081, 1.443]
TW	+0.044 [+0.011, +0.076]	1.146 [0.973, 1.350]	+0.011 [-0.025, +0.047]	1.047 [0.904, 1.214]
US	+0.024 [-0.009, +0.057]	1.039 [0.874, 1.234]	-0.020 [-0.057, +0.017]	0.923 [0.794, 1.074]
VN	+0.494 [+0.460, +0.529]	7.901 [6.771, 9.219]	+0.081 [+0.045, +0.118]	1.386 [1.197, 1.604]
ZA	+0.327 [+0.293, +0.362]	4.006 [3.438, 4.667]	+0.060 [+0.024, +0.096]	1.273 [1.101, 1.472]

Each row reports the country-specific IAP × Sports interaction estimate and its 95% confidence interval. DDP is the additive interaction (null = 0); RoR is the multiplicative interaction (null = 1). The table reports the complete national-market estimates rather than cross-country means.

Complete cross-market and temporal synthesis

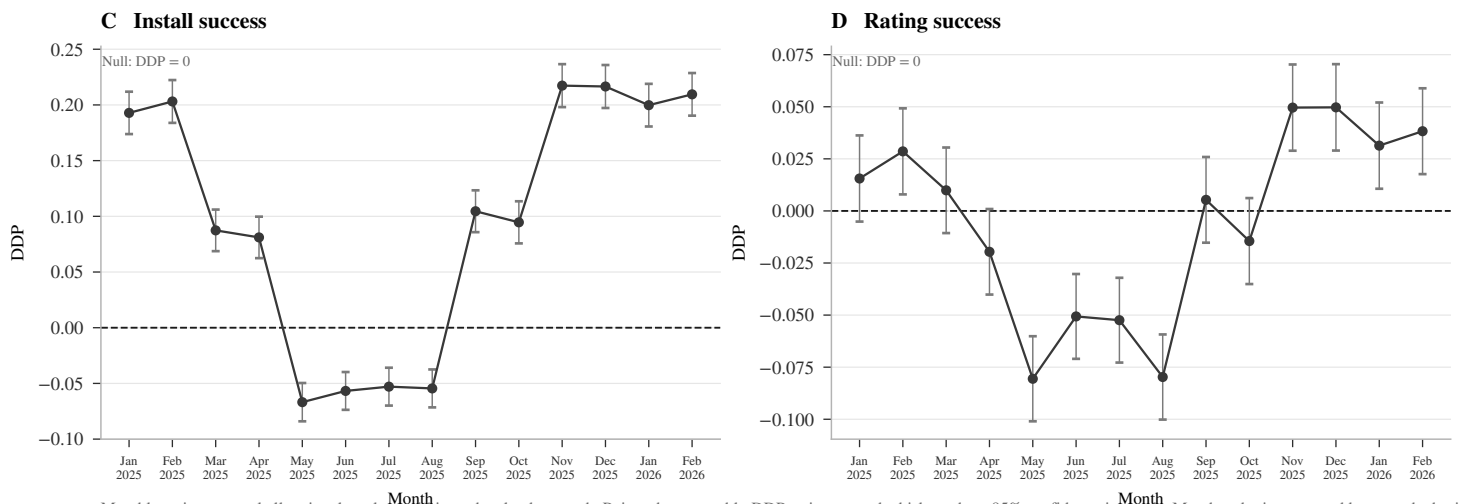
IAP × Sports interaction | Install success and rating success

Cross-market interaction — all national markets



Countries are ordered independently by estimated DDP within each outcome. Bars show the interaction estimate; whiskers show 95% confidence intervals.

Temporal interaction — all 14 pooled months



Monthly estimates pool all national markets within each calendar month. Points show monthly DDP estimates and whiskers show 95% confidence intervals. Month order is preserved because the horizontal axis represents time.

Reading the figure

DDP = 0 is the additive no-interaction reference. Values below zero indicate that the IAP-associated probability difference is more negative for Sports than for non-Sports games; values above zero indicate the reverse. The corresponding multiplicative evidence is retained in the complete numerical synthesis on the preceding page using RoR and log(RoR).